Cuong Bui

San Francisco, CA ✻ *E-mail:* cb.626@berkeley.edu ✻ *Telephone number:* 714-823-0014 ✻ *LinkedIn:* cb626

Berkeley-trained physicist with experience in machine learning, scientific computing, and software development. Passionate about applying computational methods to solve engineering and scientific challenges in mission-driven technologies.

Work Experience

Software Product Developer - Omnium LLC **January 2026 - April 2026**

- Developed a GitHub-based application launchpad that replaced existing internal tooling workflows and centralized deployment, version control, and distribution of internal analytical tools.
- Engineered a desktop application integrating waterfall chart analytics into an intuitive interface, streamlining internal analysis workflows.

Data Scientist Intern - Omnium LLC **May 2025 - December 2025**

- Developed Python- and Excel-based analytical models to evaluate pricing scenarios, forecast demand, and identify competitive market trends using large-scale consumer data.
- Designed a machine learning framework for detecting outliers in time-series pricing models using DBSCAN and Isolation Forest algorithms.
- Integrated the outlier detection framework into a production application, enabling company-wide access to automated anomaly detection.

Student Researcher - Physics Department, UC Berkeley **January 2025 - May 2025**

- Conducted independent research developing and training a convolutional neural network (CNN) for classification of particle-detector pulse signals in noisy time-series data.
- Presented deep learning research findings at collaboration meetings and American Physical Society (APS) conferences.

Undergraduate Student Instructor - Astronomy Department, UC Berkeley **August 2024 - December 2024**

- Designed and taught weekly discussion sections, quizzes, and office hours for an introductory astronomy course serving over 700 students.

Department Student Advisor - Astronomy Department, UC Berkeley **January 2024 - December 2024**

- Mentored undergraduate astronomy students on academic planning, research opportunities, and course selection in collaboration with faculty advisors.

Projects

Undergraduate Honors Thesis (High Honors in Physics) **August 2023 - December 2024**

"Using Convolutional Neural Networks for Event Pileup Discrimination in CUPID"

- Generated and analyzed over 100,000 simulated particle-detector signals to train and evaluate CNN models.
- Evaluated convolutional neural network architectures for event pileup discrimination, improving detector background noise rejection in the CUPID experiment.

Intro to Game Development Decal Lead Facilitator **January 2024 - December 2024**

- Developed curriculum and mentored six student teams through semester-long game design projects using Roblox Studio, accredited as a UC Berkeley student-run elective course.

Education

B.A. in Astrophysics and High Honors in Physics - UC Berkeley **June 2020 - December 2024**

GPA: 3.86/4.00

Relevant Coursework: Multivariable Calculus, Linear Algebra and Differential Equations, Computer Simulations, Analytic Mechanics, Data Structures and Algorithms, Bayesian Data Analysis and Machine Learning, Concepts of Probability.

Skills

Programming: Python, Java, R

Machine Learning: TensorFlow, Scikit-learn, CNNs, Regression, Time-Series Analysis, DBSCAN, Isolation Forest

Software: Git, GitHub, Jupyter, VS Code, Excel, LaTeX, NumPy, Pandas, Matplotlib

Analysis: Statistical Modeling, Data Visualization, Signal Processing

Interests and Extracurriculars

Colorguard Performer **2019 - 2026**

- Pacific Crest Drum and Bugle Corps (2019, 2021), Bluecoats Drum and Bugle Corps (2022, 2024), and Vox Artium Winter Guard (2026) performer.
- Performed with nationally ranked ensembles for collective audiences exceeding 300,000 while balancing rigorous academics and research commitments.